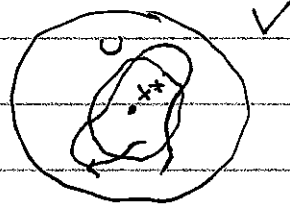


Stability of a fixed pt. x^* :

$$\dot{x} = f(x), x \in \mathbb{R}^n, f(x^*) = 0$$

x^* is Lyapunov stable if all trajectories that start sufficiently close to it, remain close to it for all ^{future} time.

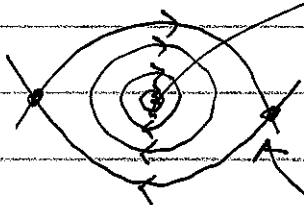
Given any neighborhood V of x^* I can find a neighborhood $U \subset V$ s.t. if $x(0) \in U$, $x(t) \in V \quad \forall t > 0$.



If x^* is Lyapunov stable and nearby trajectories converge to x^* as $t \rightarrow \infty$, then x^* is asymptotically stable.

If x^* isn't stable, it's unstable.

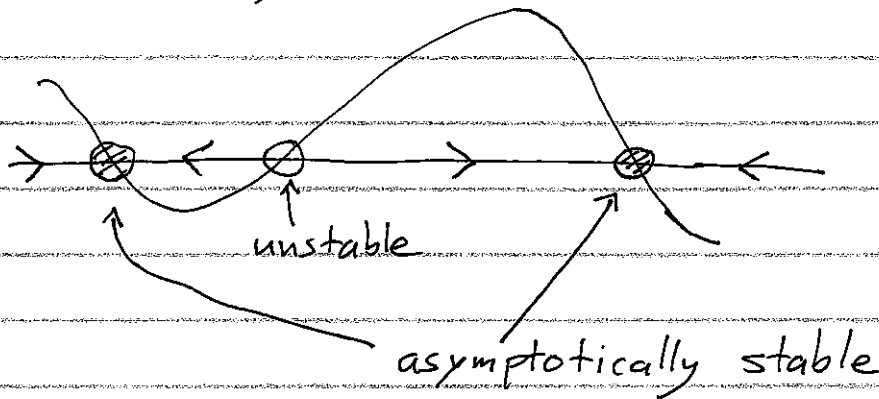
Pendulum:



Lyapunov stable, not asymptotically stable

unstable

$$\dot{x} = f(x), \quad x \in \mathbb{R} :$$



An example for which x^* is Lyapunov stable, but not asymptotically stable?

Linear stability: $x = x^* + y$ $f(x^*) = 0$

$$\frac{dx}{dt} = \frac{dy}{dt} = f(x^* + y) = f(x^*) + f'(x^*)y + \dots$$

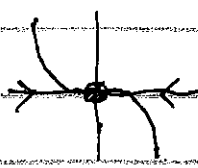
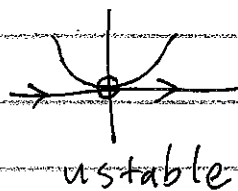
$$\frac{dy}{dt} = \underbrace{f'(x^*)}_{\mu} y \Rightarrow y(t) = e^{\mu t} y_0$$

$\mu > 0 \Rightarrow$ linearly unstable $\Rightarrow$ unstable

$\mu < 0 \Rightarrow$ linearly stable $\Rightarrow$ asym. stable

can show

$\mu = 0$? linear stability inconclusive



asymptotically stable

Model of global energy balance & Earth's temp.
Demonstrates

- modeling positive feedback (ice albedo feedback)
- bistability
- bifurcations
- hysteresis

Let $T(t)$ = globally averaged, annual mean temperature of Earth's surface in Kelvin

20th century average $\approx 57^\circ\text{F} \approx 287\text{ K}$
(2016 : 288 K)

Energy balance model:

$$C \frac{dT}{dt} = F_{\text{in}} - F_{\text{out}} \quad \leftarrow \text{units of } \text{W/m}^2$$

specific "heat capacity" of Earth

Amount of thermal energy required to raise temp of 1 m^2 surface area of Earth by 1°C (or 1 K , same)

$\sim 1 - 100 \text{ (W.yrs/m}^2\text{ }^\circ\text{C)}$ orders of magnitude

F_{in} = absorbed incoming solar radiation

incoming radiation available to warm planet:

$$\begin{aligned} \text{cross-sectional area} \rightarrow \left(\frac{\pi r^2}{4\pi r^2} \right) \cdot 1368 \text{ W/m}^2 &= \frac{1368}{4} \text{ W/m}^2 \\ &\quad \swarrow \text{"solar constant"} \\ &= 342 \text{ W/m}^2 \\ &\quad \nwarrow \text{surface area} \end{aligned}$$

But some is reflected

α = albedo $\in [0, 1]$ = fraction that's reflected

$1 - \alpha$ = co-albedo = fraction that's absorbed

$$F_{in} = (1 - \alpha) Q, \quad Q = 342 \text{ W/m}^2$$

$\alpha = ?$, depends on Earth's surface

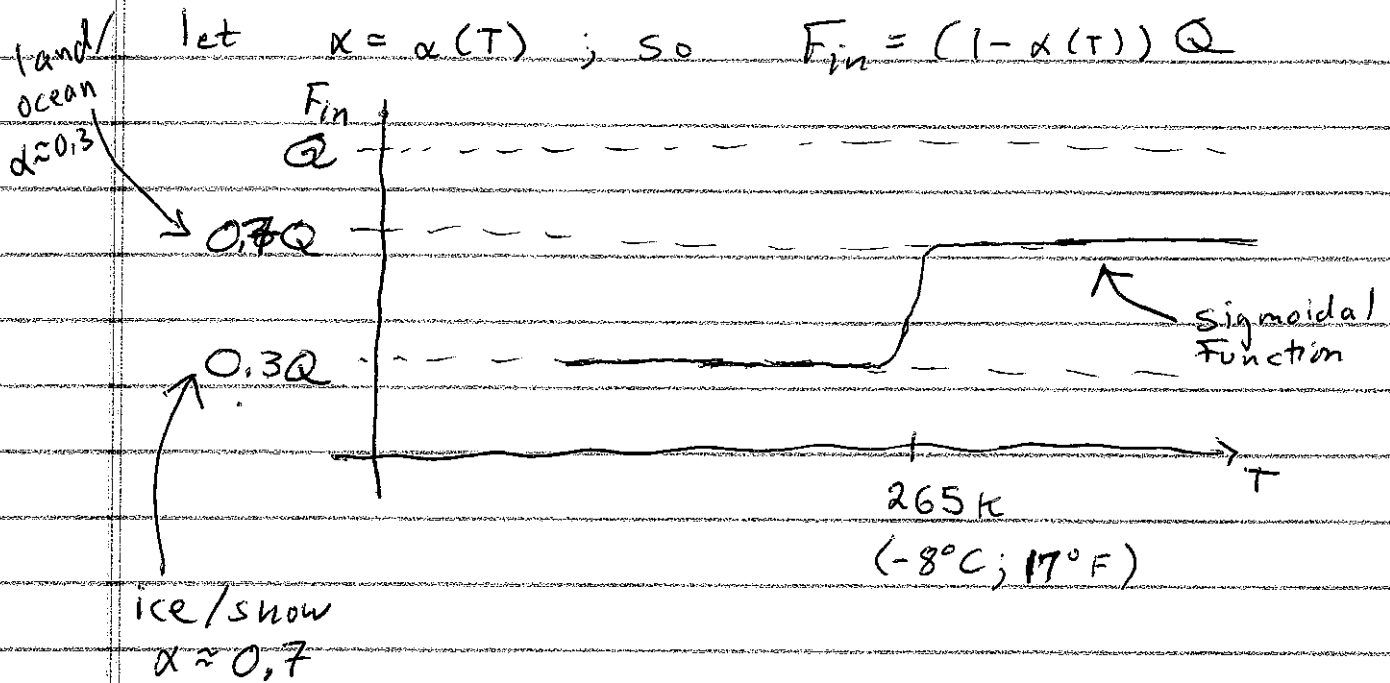
α_i = snow-ice $\alpha >$ ocean-land $\alpha = \alpha_o$

"ice-albedo feedback"

Temp $\uparrow$, then snow/ice melt, then $\alpha \downarrow$ and $F_{in} \uparrow$, further increase in T

Temp $\downarrow$, then amount of snow/ice increases, then $\alpha \uparrow$ and $F_{in} \downarrow$, further decrease in T

Lecture 3 p.5



$F_{out} = \sigma T^4$ is Earth is a perfect black body at temperature T

Stefan-Boltzmann constant = $5.67 \times 10^{-8} \frac{W}{m^2 K^4}$

$$C \frac{dT}{dt} = (1 - \alpha(T)) Q - \sigma T^4$$

equilibrium temperature: $F_{in}(T^*) = F_{out}(T^*)$

$$T^* = \left[\frac{(1 - \alpha) Q}{\sigma} \right]^{1/4} \approx \begin{cases} 255 K & (\alpha = 0.3) \\ 206 K & (\alpha = 0.7) \end{cases}$$

← wrong α

← yes!

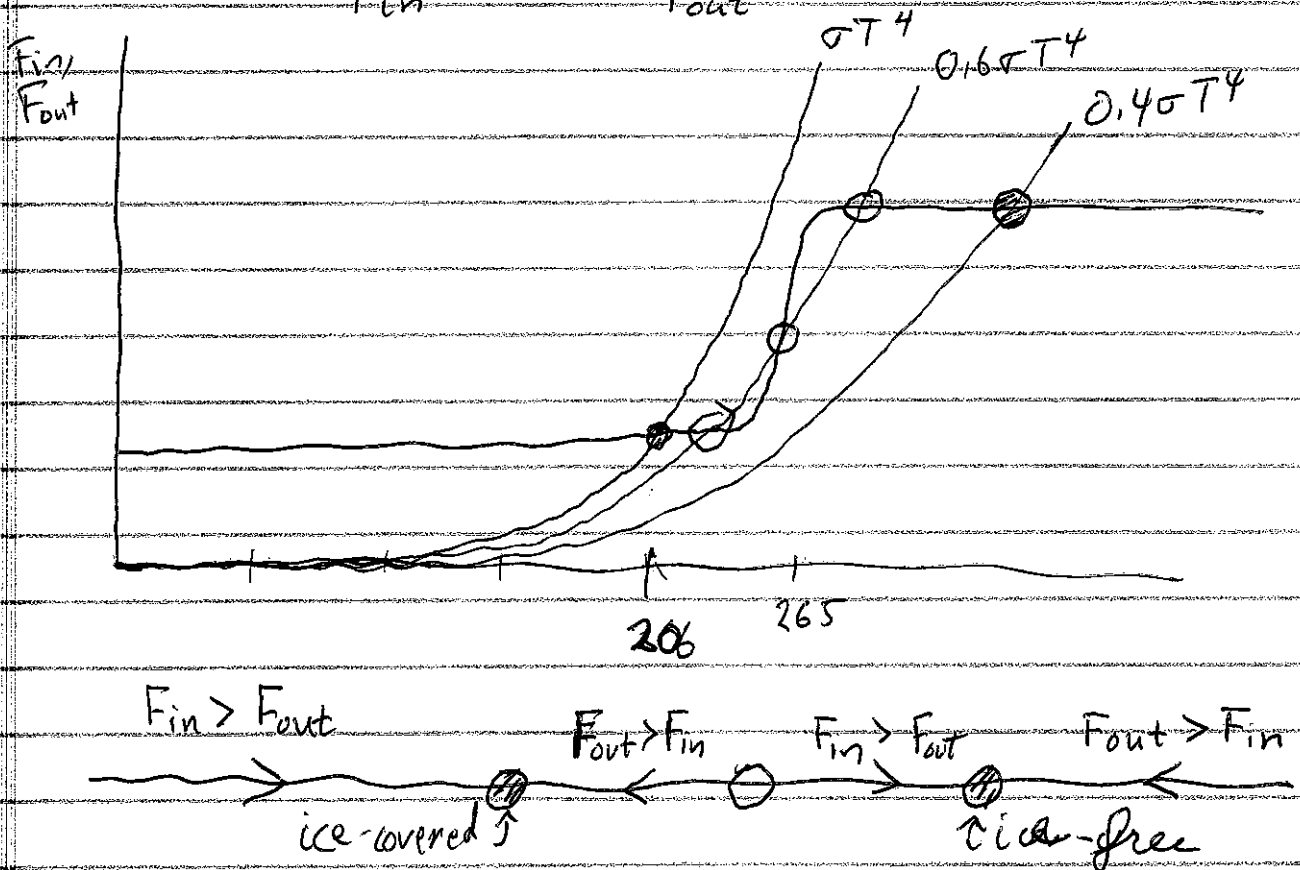
Modification of the model to include a Greenhouse gas effect

$$F_{\text{out}} = \epsilon \sigma T^4$$

"emissivity" $\epsilon \leq 1$; $\epsilon = 1 = \text{perfect blackbody}$

assume that as Greenhouse gases increase, ϵ decreases

$$C \frac{dT}{dt} = \underbrace{(1 - \alpha(T)) Q}_{F_{\text{in}}} - \underbrace{\epsilon \sigma T^4}_{F_{\text{out}}}$$



Lecture 4 p.1

$$C \frac{dT}{dt} = F_{in} - F_{out}$$

$$\epsilon \in [0, 1]$$

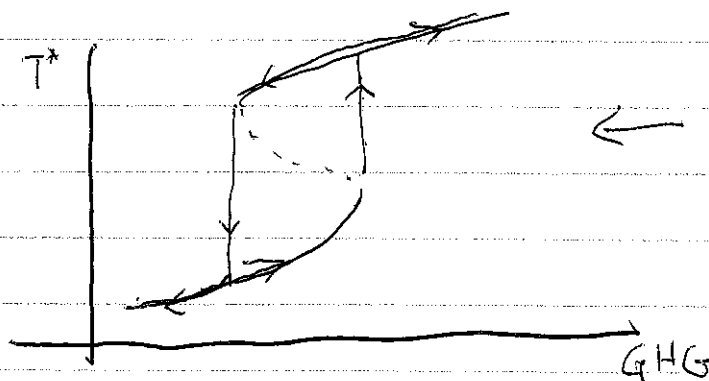
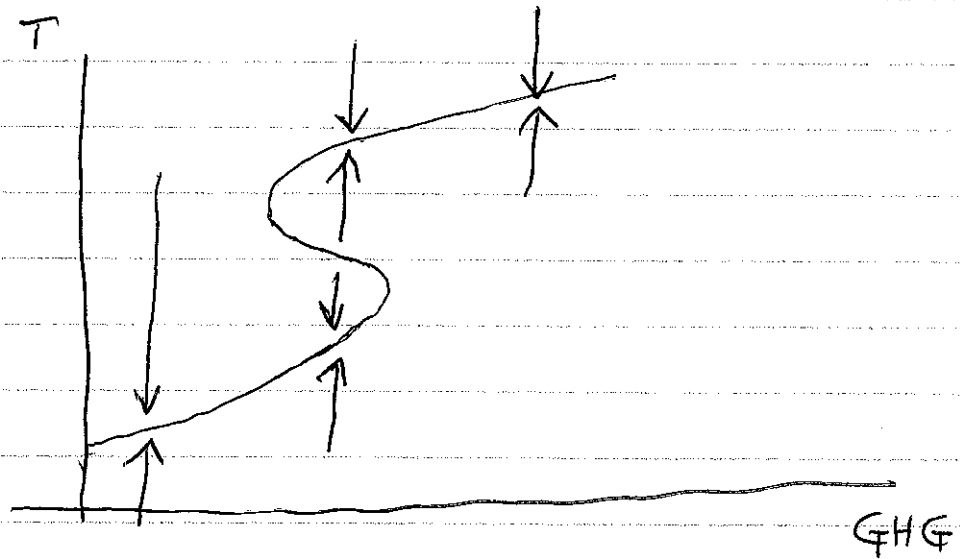
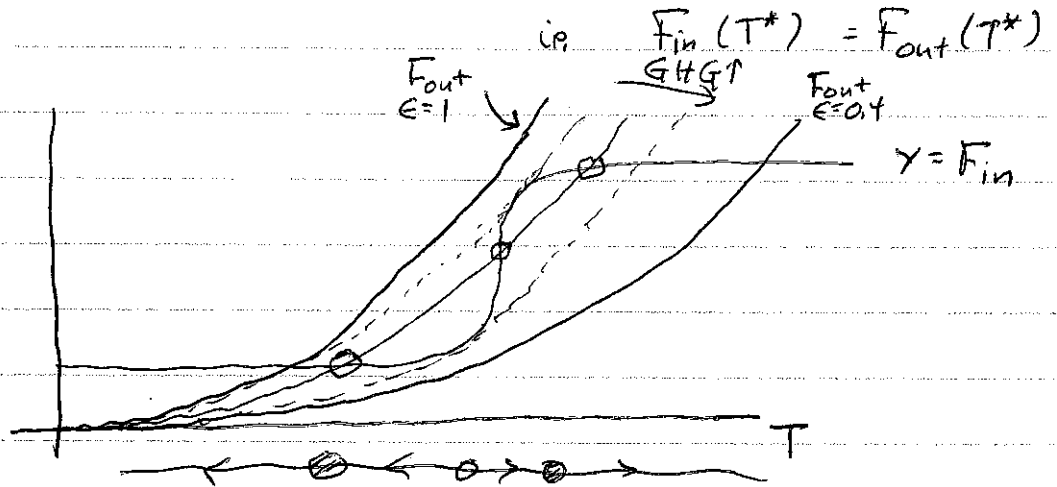
$$F_{in} = (1 - \alpha(T)) Q$$

$$F_{out} = \epsilon \sigma T^4$$

equilibria T^* satisfy

$$F_{in}(T^*) - F_{out}(T^*) = 0$$

$$F_{in}(T^*) = F_{out}(T^*)$$



← Bifurcation Diagram demonstrating hysteresis loop